

RIGIDITY OF THE EULER–ARNOLD FORM UNDER ENERGY-PRESERVING DEFORMATION

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ABSTRACT. The Euler nonlinearity is not merely a bilinear operator obeying an energy identity: by Arnold’s theorem it is the Euler–Arnold operator $\dot{m} = \text{ad}_{A^{-1}m}^* m$ of a Lie algebra with inertia operator A . We ask how far that form can be deformed while preserving the energy identity, and find it rigid. Two classes of deformation are treated: triad reweightings of the Euler vertex, and vertices twisted by scalar Fourier multipliers on the three legs. Tao’s operator lies in neither, and is reached by a separate argument. On $\text{SDiff}(\mathbb{T}^3)$ the energy-preserving triad reweightings of Euler that remain Euler–Arnold are exactly the global rescalings, and the family does not reopen when anisotropic, matrix-valued inertia operators are allowed. The two-dimensional picture is different, and is what gives the three-dimensional statement its content: on Zeitlin’s $\mathfrak{su}(N)$ truncation of $\text{SDiff}(\mathbb{T}^2)$ the L^2 pairing is ad-invariant, every Euler–Arnold vertex on a triad is a scalar multiple of every other, and the corresponding family is two-dimensional. Its non-trivial direction is the Camassa–Holm inertia operator: where the structure admits a deformation at all, what it singles out is a recognisable geometric object rather than an artifact of the truncation.

The two-dimensional mechanism is a local-to-global collinearity statement: the deformation condition says exactly that $m \mapsto (a_m, a'_m)$ carries every Fourier triad to three collinear points of the plane, and one then propagates a single line across the lattice. In three dimensions the L^2 pairing is not ad-invariant, so the per-triad vertex space has dimension two rather than one, and the ratio of the symbol at two legs becomes readable pointwise from the vertex. For vertices twisted by scalar Fourier multipliers on the three legs we obtain a complete characterisation: energy conservation forces the first and third symbols to be proportional, and the Euler–Arnold property then holds if and only if the product of the first two is constant. This does not survive averaging, and we give two explicit counterexamples: a positively weighted average of twists, none of which conserves energy, conserves it exactly when a frame condition $\sum_j c_j v_j v_j^\top \propto I$ holds; and an average of two Euler–Arnold vertices with distinct symbols is again Euler–Arnold whenever the logarithm of their ratio is a parity function.

We then apply this to Tao’s averaged Navier–Stokes operator, for which finite-time blowup occurs. A mode triple of his cascade is compatible with the Euler bracket only if the underlying frequencies can sum to zero, and because all of his profiles live in one thin annulus this fails whenever a triple carries fewer than three distinct species. Twelve of the sixteen non-zero structure constants of his Table 1 carry fewer than three, and among them is every coupling that transfers energy between scales: the part of his cascade that the Euler bracket can accommodate acts within a single scale and cannot cascade at all. A separate range obstruction covers inertia operators that are not translation covariant. This excludes his operator from the Euler–Arnold class. It does not exclude the existence of an Euler–Arnold operator that blows up: one exists, by a second theorem of Tao, whose inertia operator is inhomogeneous and of the same order as Euler’s. What the results here bear on is the translation-covariant case, and Section 7 poses the surviving question for isotropic symbols. It is narrow for a structural reason rather than a technical one: the known routes to blowup for such a geodesic flow either break the homogeneity of the metric, as Tao’s does, or drive the Jacobian of the Lagrangian flow map to zero, which volume preservation forbids.

Every statement is proved. Three hypotheses are finite conditions on a symbol or on a truncation; these are verified for each case used, by an accompanying script, rather than established in general.

1. INTRODUCTION

1.1. **The question.** Write the Navier–Stokes equation on $\mathbb{R}^3$ or $\mathbb{T}^3$ as

$$\partial_t u = \Delta u + B(u, u), \quad \operatorname{div} u = 0, \quad (1)$$

with B the Euler bilinear operator. Tao [19] constructs an averaged operator $\tilde{B}$, obtained from B by composing with order-zero Fourier multipliers, rotations and dilations and then averaging, which obeys the same cancellation law $\langle \tilde{B}(u, u), u \rangle = 0$ and the same scaling, and for which (1) blows up in finite time. The moral usually drawn is that any proof of global regularity must use more about B than its energy identity and its scaling.

This note asks what "more" can mean, and takes one specific answer seriously: B is not merely a bilinear operator with a cancellation property, it is the Euler–Arnold operator of a Lie algebra. Arnold’s theorem [1] realises the incompressible Euler equation as geodesic flow on the group of volume-preserving diffeomorphisms; in Lie–Poisson form,

$$\dot{m} = \operatorname{ad}_{A^{-1}m}^* m, \quad (2)$$

where $\mathfrak{g}$ is the Lie algebra of divergence-free vector fields, $m \in \mathfrak{g}^*$, and $A: \mathfrak{g} \rightarrow \mathfrak{g}^*$ is the inertia operator of the metric. Euler is the case $A = \operatorname{id}$. We ask:

Which energy-preserving deformations of B are of the form (2)?

1.2. **What is shown.** Sections 2–3 treat the two-dimensional model problem, Zeitlin’s $\mathfrak{su}(N)$ truncation of $\operatorname{SDiff}(\mathbb{T}^2)$. There the Euler–Arnold vertex on a Fourier triad is a single scalar, and the question of which triad reweightings preserve the Euler–Arnold form becomes a *linear* system in the symbol (Theorem 3.3). Its solution space always contains the two-dimensional gauge $\operatorname{span}\{a, 1\}$, and Theorem 3.8 shows it is never larger, under a finite condition on the symbol alone which we verify. The mechanism is a local-to-global collinearity statement: the linear system says exactly that the map $m \mapsto (a_m, a'_m)$ carries every Fourier triad to three collinear points of the plane (Lemma 3.5), and one then propagates a single line across the lattice. The non-trivial direction of the resulting gauge is exactly the Camassa–Holm inertia operator (Corollary 3.11) — a recognisable geometric object, not an artifact of the truncation.

Section 4 sets up three dimensions. The essential structural difference is Lemma 4.1: on divergence-free vector fields the L^2 pairing is *not* ad-invariant, so the structure form is anti-symmetric in its first and third slots only, not totally. This is exactly why $\mathfrak{su}(N)$ with $A = \operatorname{id}$ is trivial while three-dimensional Euler with $A = \operatorname{id}$ is the whole problem, and it makes the per-triad vertex space two-dimensional instead of one-dimensional.

Section 5 contains the three-dimensional results. The ratio a_l/a_k is readable pointwise from the vertex (Theorem 5.1), so the obstruction is linear in $\log a$; the gauge of Theorem 3.3 collapses from two dimensions to one (Theorem 5.2); allowing anisotropic, matrix-valued inertia operators does not improve it (Theorem 5.9); and for vertices twisted by scalar multipliers on the three legs we give a complete characterisation (Theorem 5.5). That characterisation fails for averages, in both of its halves: Theorems 5.12 and 5.14 give explicit positively weighted averages which conserve energy, respectively remain Euler–Arnold, although no term-by-term criterion predicts it. Remark 5.4 compares the two dimensions and explains why the three-dimensional collapse is the stronger statement, and why it does not combine with the two-dimensional one into a single argument.

Section 6 applies this to Tao’s operator. Theorem 6.2 is a range obstruction valid for every positive definite A and independent of the structure constants. Theorem 6.3 is sharper and is the main point of the section: a cascade term is compatible with the Euler bracket only if its frequencies can sum to zero, which fails whenever it carries fewer than three distinct species,

and twelve of Tao’s sixteen do. Corollary 6.5 sharpens this: the twelve include every coupling that moves energy between scales, so what survives is single-scale and provably bounded.

1.3. What is not shown. None of this bears on the regularity of genuine Navier–Stokes. Section 6 excludes Tao’s operator from the Euler–Arnold class; it does not show that no Euler–Arnold operator blows up, and that stronger statement is in fact false [20]. Theorem 5.14 was a reason for caution about it, and Section 7 sets out what survives: the inertia operators known to blow up are inhomogeneous, and the translation-covariant case in which our symbol arguments are stated remains open. Throughout, the Lie bracket is held fixed and only the inertia operator is varied. Every statement below is proved. Three hypotheses — reachability (Definition 3.7), transversality (Definition 5.8) and connectedness of the triangle and triad graphs — are finite conditions on a symbol or on a truncation, which we verify for each case used rather than establish in general; a script accompanying the paper checks them all.

2. PRELIMINARIES

Let $\mathfrak{g}$ be a Lie algebra with an inner product $\langle \cdot, \cdot \rangle$ and let $A: \mathfrak{g} \rightarrow \mathfrak{g}$ be symmetric and positive definite. Define ad^* by $\langle \text{ad}_x^* y, z \rangle = \langle y, [x, z] \rangle$. The Euler–Arnold equation is (2), and its conserved energy is $\frac{1}{2} \langle m, A^{-1} m \rangle$.

It is convenient to pass to energy-normalised variables $X = A^{1/2} m$, so that the conserved quantity becomes the plain norm $\frac{1}{2} \|X\|^2$; this is the normalisation in which Tao’s cancellation condition $\langle \tilde{B}(u, u), u \rangle = 0$ is stated. In these variables (2) reads $\dot{X} = B_A(X, X)$ with

$$B_A(X, Y) = \frac{1}{2} \left(A^{-1/2} \text{ad}_{A^{-1/2} X}^* (A^{1/2} Y) + A^{-1/2} \text{ad}_{A^{-1/2} Y}^* (A^{1/2} X) \right), \quad (3)$$

and we write $T_A(X, Y, Z) := \langle B_A(X, Y), Z \rangle$ for the associated trilinear form, symmetric in its first two arguments. Set

$$S(x, y, z) := \langle y, [x, z] \rangle, \quad (4)$$

so that $T_A(X, Y, Z) = \frac{1}{2} (S(A^{-1/2} X, A^{1/2} Y, A^{-1/2} Z) + S(A^{-1/2} Y, A^{1/2} X, A^{-1/2} Z))$.

Lemma 2.1. *S is antisymmetric in its first and third arguments: $S(x, y, z) = -S(z, y, x)$. Consequently $T_A(X, X, X) = 0$ for every A , that is, every Euler–Arnold system conserves energy.*

Proof. $S(x, y, z) = \langle y, [x, z] \rangle = -\langle y, [z, x] \rangle = -S(z, y, x)$. Taking $X = Y = Z$ in (3) gives $T_A(X, X, X) = S(A^{-1/2} X, A^{1/2} X, A^{-1/2} X)$, whose first and third arguments coincide, so it vanishes. $\square$

3. THE ZEITLIN TRUNCATION

3.1. Setup. Let N be odd, $\omega = e^{2\pi i/N}$, $s = 2^{-1} \bmod N$, and let $g = \text{diag}(1, \omega, \dots, \omega^{N-1})$ and h be the cyclic shift, so $gh = \omega hg$. For $m \in \mathbb{Z}_N^2 \setminus \{0\}$ put

$$J_m := \omega^{sm_1 m_2} g^{m_1} h^{m_2},$$

a basis of $\mathfrak{sl}(N, \mathbb{C})$. This is the Fairlie–Fletcher–Zachos sine algebra, and as $N \rightarrow \infty$ its structure constants converge to the Poisson bracket $\{e_m, e_n\} = (m \times n) e_{m+n}$ on the torus; this is Zeitlin’s truncation [21] of $\text{SDiff}(\mathbb{T}^2)$. Equip $\mathfrak{sl}(N, \mathbb{C})$ with the invariant form $\langle X, Y \rangle = \text{tr}(XY)$ and set

$$F(m, n, p) := \text{tr}([J_m, J_n] J_p), \quad (5)$$

supported on $m + n + p \equiv 0$. Since $\text{tr}([x, y]z)$ is totally antisymmetric for any Lie algebra with an invariant trace form, so is F ; in particular F is invariant under cyclic permutation of a triangle $\{m, n, p\}$. That single fact is what makes the cancellation condition below a per-triangle condition. Call a triangle *degenerate* if F vanishes on it.

Let A be diagonal in this basis, $AJ_m = a_m J_m$, with $a_{-m} = a_m > 0$; diagonality is forced if one asks A to commute with the torus translations $J_m \mapsto e^{i\theta \cdot m} J_m$.

Proposition 3.1. *For $m + n + p \equiv 0$,*

$$T_a(m, n, p) = \frac{1}{2} F(m, n, p) \frac{a_m - a_n}{\sqrt{a_m a_n a_p}}. \quad (6)$$

Moreover $T_a(m, n, p) + T_a(n, p, m) + T_a(p, m, n) = 0$ identically.

Proof. For $\mathfrak{sl}(N, \mathbb{C})$ with the invariant form one has $\text{ad}_x^* y = -[x, y]$, so $\langle A^{-1/2}[x, y], z \rangle = f(x, y, A^{-1/2}z)$ with $f(x, y, z) = \text{tr}([x, y]z)$ totally antisymmetric. Evaluating (3) on basis elements J_m, J_n, J_p gives

$$T_a(m, n, p) = -\frac{1}{2} F(m, n, p) \left(a_m^{-1/2} a_n^{1/2} a_p^{-1/2} - a_n^{-1/2} a_m^{1/2} a_p^{-1/2} \right),$$

the sign in the second term coming from $f(J_n, J_m, J_p) = -F(m, n, p)$. Combining the fractions yields (6). For the second claim, F is constant on the cyclic orbit of $\{m, n, p\}$ and the denominators agree, so the three terms are $\frac{F}{2\sqrt{a_m a_n a_p}}$ times $(a_m - a_n) + (a_n - a_p) + (a_p - a_m) = 0$. $\square$

3.2. Triad reweightings. Tao's averaging reweights interactions while preserving the triad cancellation. Its finite-dimensional counterpart is the following.

Definition 3.2. A *triad reweighting* of T_a is $\tilde{T} := \mu_\tau T_a$ for a function μ on triangles with values in $\mathbb{R} \setminus \{0\}$.

By Proposition 3.1 any reweighting again conserves energy, is translation covariant and has the symmetry of a vertex. The question is which reweightings remain Euler–Arnold.

Theorem 3.3. *Let a be a positive symbol and μ a reweighting. Then $\mu T_a = T_{a'}$ for a positive symbol a' if and only if a' satisfies, on every non-degenerate triangle $\{m, n, p\}$,*

$$(a'_m - a'_n)(a_n - a_p) = (a'_n - a'_p)(a_m - a_n) \quad (7)$$

together with its cyclic images. The reweighting is then determined by a' through $\mu_\tau = \frac{a'_m - a'_n}{a_m - a_n} \sqrt{\frac{a_m a_n a_p}{a'_m a'_n a'_p}}$. In particular the condition on a' is linear.

Proof. Fix a non-degenerate triangle $\tau = \{m, n, p\}$. By (6), matching $\mu_\tau T_a = T_{a'}$ on the pair (m, n) gives

$$\mu_\tau \frac{a_m - a_n}{\sqrt{a_m a_n a_p}} = \frac{a'_m - a'_n}{\sqrt{a'_m a'_n a'_p}},$$

the common factor $F(m, n, p) \neq 0$ cancelling. The two square roots do not depend on which pair of τ is taken, so writing ρ_τ for the common value of $(a'_m - a'_n)/(a_m - a_n)$ over the three pairs of τ we obtain $\rho_\tau = \mu_\tau \sqrt{a'_m a'_n a'_p} / \sqrt{a_m a_n a_p}$, which is the stated formula for μ_τ , and the equality of the three ratios is exactly (7) and its cyclic images. Conversely, given a' satisfying (7), defining μ_τ by that formula reproduces $T_{a'}$. $\square$

Write $L(a)$ for the linear map on symbols defined by the left-hand sides of (7) over all non-degenerate triangles.

Lemma 3.4. $\text{span}\{a, \mathbf{1}\} \subseteq \ker L(a)$.

Proof. For $a' = \mathbf{1}$ all differences vanish. For $a' = a$ both sides of (7) equal $(a_m - a_n)(a_n - a_p)$. The condition is linear in a' . $\square$

3.3. The kernel. The condition (7) has a geometric reading which determines the kernel. For a symbol a and a candidate a' define

$$\Phi: \Lambda \longrightarrow \mathbb{R}^2, \quad \Phi(m) := (a_m, a'_m), \quad (8)$$

where $\Lambda := \mathbb{Z}_N^2 \setminus \{0\}$. Since symbols are even, $\Phi(-m) = \Phi(m)$, so Φ factors through the set of $\pm$ classes.

Lemma 3.5 (Collinearity). *$a' \in \ker L(a)$ if and only if for every non-degenerate triangle $\{m, n, p\}$ the three points $\Phi(m), \Phi(n), \Phi(p)$ are collinear in $\mathbb{R}^2$.*

Proof. Condition (7) together with its two cyclic images states precisely that all 2×2 minors of

$$\begin{pmatrix} a_m - a_n & a_n - a_p & a_p - a_m \\ a'_m - a'_n & a'_n - a'_p & a'_p - a'_m \end{pmatrix}$$

vanish, that is, that this matrix has rank at most one. Its columns are the three difference vectors $\Phi(m) - \Phi(n)$, $\Phi(n) - \Phi(p)$, $\Phi(p) - \Phi(m)$, written in coordinates; rank at most one says exactly that these are pairwise parallel, which is collinearity of the three points. (If the first row vanishes, so that $a_m = a_n = a_p$, the three points lie on a vertical line and both statements hold.) $\square$

Corollary 3.6 (Level sets). *Let $\{m, n, p\}$ be non-degenerate with $a_m = a_n$ and $a_n \neq a_p$. Then every $a' \in \ker L(a)$ satisfies $a'_m = a'_n$.*

Proof. $\Phi(m)$ and $\Phi(n)$ lie on the vertical line $x = a_m$, while $\Phi(p)$ does not. Three collinear points, two of which lie on a vertical line not containing the third, force the first two to coincide. $\square$

By Lemma 3.5, determining $\ker L(a)$ amounts to a local-to-global question: if Φ carries every triangle to a collinear triple, is $\Phi(\Lambda)$ contained in a single line? This is false without a hypothesis on a — if a is constant then every condition is vacuous — and the following finite condition, which involves a alone and not a' , is what the argument needs. Write $m \times n := m_1 n_2 - m_2 n_1$, so that a triangle is non-degenerate exactly when its vertices are pairwise non-parallel mod N .

Definition 3.7 (Reachability). A symbol a is *reachable* if there are $m_0, n_0 \in \Lambda$ with $m_0 \times n_0 \not\equiv 0$ and $a_{m_0} \neq a_{n_0}$ such that the following closure exhausts Λ : start with $T = \{m_0, n_0\}$, and whenever $u, v \in T$ satisfy $u \times v \not\equiv 0$ and $a_u \neq a_v$, adjoin $u + v$ and $u - v$.

Theorem 3.8. *Let N be odd and let a be a reachable symbol. Then*

$$\ker L(a) = \text{span}\{a, \mathbf{1}\},$$

a space of dimension exactly 2.

Proof. The inclusion $\supseteq$ is Lemma 3.4, and $a, \mathbf{1}$ are independent because $a_{m_0} \neq a_{n_0}$. For $\subseteq$, let $a' \in \ker L(a)$ and let Φ be as in (8).

Fix m_0, n_0 as in Definition 3.7. Since $a_{m_0} \neq a_{n_0}$ the points $\Phi(m_0)$ and $\Phi(n_0)$ are distinct; let ℓ be the line through them, and put

$$S := \{x \in \Lambda : \Phi(x) \in \ell\} \supseteq \{m_0, n_0\}.$$

Propagation. Suppose $u, v \in S$ with $u \times v \not\equiv 0$ and $a_u \neq a_v$. We claim $u + v \in S$ and $u - v \in S$. First, $u + v \neq 0$, since $v = -u$ would be parallel to u ; so $\{u, v, -(u + v)\}$ is a triangle, and it is non-degenerate because $u \times v \not\equiv 0$. By Lemma 3.5 the points $\Phi(u), \Phi(v), \Phi(-(u + v))$ are collinear, and $\Phi(u) \neq \Phi(v)$ because $a_u \neq a_v$; hence the line through them is ℓ , and $\Phi(u + v) = \Phi(-(u + v)) \in \ell$. For $u - v$, apply the same argument to the pair $(u, -v)$, which is again non-parallel, has $a_{-v} = a_v \neq a_u$, and satisfies $\Phi(-v) = \Phi(v) \in \ell$.

Exhaustion. The propagation step is precisely the closure rule of Definition 3.7, and S contains the initial pair $\{m_0, n_0\}$ and is closed under it. Since a is reachable, that closure exhausts Λ , so $S = \Lambda$, that is, $\Phi(\Lambda) \subseteq \ell$.

Conclusion. The points $\Phi(m_0), \Phi(n_0) \in \ell$ have distinct first coordinates, so ℓ is not vertical and may be written $\ell = \{(t, \alpha t + \beta) : t \in \mathbb{R}\}$. Then $a'_m = \alpha a_m + \beta$ for every m , i.e. $a' \in \text{span}\{a, \mathbf{1}\}$. $\square$

Reachability is not an artificial hypothesis: it holds for generic symbols, by a clean argument, and we have verified it for the symbols of interest.

Proposition 3.9. *Let N be an odd prime and suppose a separates non-parallel classes, that is $a_m \neq a_n$ whenever $m \times n \not\equiv 0$. Then a is reachable.*

Proof. Every non-parallel pair in T is then usable, so T is closed under $u \pm v$ for all non-parallel $u, v \in T$. Choose any m_0, n_0 with $m_0 \times n_0 \not\equiv 0$; as N is prime, $m_0 \times n_0$ is a unit and $\{m_0, n_0\}$ is a basis of $(\mathbb{Z}/N)^2$. Write $w_{j,k} := jm_0 + kn_0$.

First, $w_{j,1} \in T$ for every j : this holds for $j = 0$, and $w_{j,1} = w_{j-1,1} + m_0$ with $w_{j-1,1} \times m_0 = n_0 \times m_0 \not\equiv 0$, so the step is legitimate; note $w_{j,1} \neq 0$ because $\{m_0, n_0\}$ is a basis. Next, $w_{j,0} = jm_0 \in T$ for $j \not\equiv 0$: write $jm_0 = w_{j-1,1} + (m_0 - n_0)$, where $m_0 - n_0 \in T$ and $w_{j-1,1} \times (m_0 - n_0) = -j(m_0 \times n_0) \not\equiv 0$.

Now induct on k , the cases $k = 0$ and $k = 1$ being settled. Assume $w_{j,k} \in T$ for all j with $w_{j,k} \neq 0$, and let $k' = k \pm 1$. For $j \not\equiv 0$ write $w_{j,k'} = w_{j,k} \pm n_0$; this step is legitimate since $w_{j,k} \times n_0 = j(m_0 \times n_0) \not\equiv 0$. For $j \equiv 0$ we must reach $w_{0,k'} = k'n_0$, which is non-zero only when $k' \not\equiv 0$; then

$$k'n_0 = w_{1,k'-1} + (n_0 - m_0), \quad w_{1,k'-1} \times (n_0 - m_0) = k'(m_0 \times n_0) \not\equiv 0,$$

where $w_{1,k'-1} \in T$ by the inductive hypothesis and $n_0 - m_0 = w_{-1,1} \in T$ by the first step. Hence every $w_{j,k} \neq 0$ lies in T , and $T = \Lambda$. $\square$

Remark 3.10. Reachability holds for every $N \in \{5, 7, 9, 11, 13\}$ and each of the symbols $a_m = |m|^2$ (with m lifted to $[-(N-1)/2, (N-1)/2]^2$), the sine-Laplacian, $m_1^2 + 2.7m_2^2$, $|m|^4$, and random positive even symbols: in all twenty-five cases the closure of Definition 3.7 exhausts Λ , starting from $\{(0, 1), (1, 1)\}$ or $\{(0, 1), (1, 0)\}$. Note that of these only the random symbols separate non-parallel classes, so Proposition 3.9 does not cover the others; reachability is strictly weaker and does. Given this, Theorem 3.8 determines the kernel in all twenty-five cases; as an independent check, the identity $\ker L(a) = \text{span}\{a, \mathbf{1}\}$ was also confirmed directly by rank computation, to 1.6×10^{-15} .

Corollary 3.11. *Let a be reachable. Then the triad reweightings of T_a that remain Euler–Arnold are exactly those induced by $a' = \alpha a + \beta$ with $\alpha > 0$ and $\alpha a + \beta > 0$, that is by $A \mapsto \alpha A + \beta$. Modulo the global rescaling $\beta = 0$, which only rescales the vertex, the non-trivial direction is $a \mapsto a + \beta$, i.e. $A = -\Delta \mapsto -\Delta + \beta$: the Helmholtz, or Camassa–Holm, inertia operator.*

Proof. Immediate from Theorems 3.3 and 3.8. For $\beta = 0$ the formula for μ gives $\mu_\tau = \alpha^{-1/2}$, a constant. $\square$

Remark 3.12. The induced reweighting for $\beta \neq 0$ is genuinely non-constant: numerically μ varies by a factor 2.2 at $\beta = 1$ and 8.1 at $\beta = 5$ for $N = 9$ with $a_m = |m|^2$. So Corollary 3.11 is not vacuous. It says that the α -model is exactly the non-trivial energy-preserving triad reweighting of two-dimensional Euler that remains Euler–Arnold; the family available is one-dimensional up to scale.

Remark 3.13. The Euler–Arnold family is small inside the class of admissible reweightings. At $N = 13$ there are 4368 non-degenerate triangles, hence that many free reweighting parameters, against the two of Corollary 3.11; restricting to reweightings equivariant under $O(2, \mathbb{Z}_N)$, the symmetry group of the symbol $|m|^2$, still leaves 582.

4. THREE DIMENSIONS

4.1. The structure form. Let $\mathfrak{g} = \mathfrak{X}_{\text{vol}}(\mathbb{T}^3)$ be the divergence-free vector fields of zero mean with bracket $[u, w] = u \cdot \nabla w - w \cdot \nabla u$ and the L^2 pairing, and let S be as in (4). Euler is the case $A = \text{id}$: one checks $\text{ad}_u^* u = -\mathbb{P}(u \cdot \nabla u)$ with $\mathbb{P}$ the Leray projection, which is (1) without viscosity.

Lemma 4.1. *On $\mathfrak{X}_{\text{vol}}(\mathbb{T}^3)$ the L^2 pairing is not ad-invariant. Explicitly,*

$$\langle [u, v], w \rangle + \langle v, [u, w] \rangle = - \int_{\mathbb{T}^3} \partial_j u_i (v_j w_i + w_j v_i) dx,$$

which does not vanish identically. Consequently S is antisymmetric in slots $1 \leftrightarrow 3$, by Lemma 2.1, but not totally antisymmetric.

Proof. Expand both terms and integrate by parts using $\text{div } u = 0$; the terms $\int u \cdot \nabla (v \cdot w)$ cancel, leaving the stated expression. $\square$

This is the decisive difference from Section 3. There the analogous form was totally antisymmetric, which collapsed the per-triangle vertex to a single scalar. Here it is not, and the per-triad vertex space has dimension two. It is also why $\mathfrak{su}(N)$ with $A = \text{id}$ has trivial dynamics $\dot{m} = 0$, geodesics of a bi-invariant metric being one-parameter subgroups, while three-dimensional Euler at $A = \text{id}$ is the problem of interest.

Lemma 4.2. *Let $k + l + m = 0$ with $a \perp k$, $b \perp l$, $c \perp m$. Then*

$$S(k, l, m)[a, b, c] = i[(a \cdot b)(c \cdot l) - (b \cdot c)(a \cdot l)]. \quad (9)$$

Proof. With u, v, w the single modes $ae^{ik \cdot x}$, $be^{il \cdot x}$, $ce^{im \cdot x}$, one computes $\int v \cdot (u \cdot \nabla w) = i(a \cdot m)(b \cdot c)$ and $\int v \cdot (w \cdot \nabla u) = i(c \cdot k)(a \cdot b)$, both up to the volume factor, so $S = i[(a \cdot m)(b \cdot c) - (c \cdot k)(a \cdot b)]$. Since $a \perp k$ and $k + l + m = 0$ we have $a \cdot m = -a \cdot l$, and likewise $c \cdot k = -c \cdot l$; substituting gives (9). $\square$

4.2. Elementary tensors. On a triad, set

$$E_1 := (a \cdot b)(c \cdot l), \quad E_2 := (b \cdot c)(a \cdot l), \quad E_3 := (a \cdot c)(b \cdot m).$$

Using $a \cdot k = b \cdot l = c \cdot m = 0$ and $k + l + m = 0$, every rotation- and parity-covariant trilinear form on $k^\perp \times l^\perp \times m^\perp$ is a combination of these three: one pairs two of the vectors and contracts the third against a wavevector, and the remaining choices are related by $c \cdot k = -c \cdot l$, $a \cdot m = -a \cdot l$, $b \cdot k = -b \cdot m$. By (9), $S = i(E_1 - E_2)$. Under cyclic rotation of the triad one checks $(E_1, E_2, E_3) \mapsto (-E_2, E_3, -E_1)$.

Proposition 4.3. *Let A be a Fourier multiplier with positive even symbol a . On a triad $k + l + m = 0$, in the basis (E_1, E_2, E_3) ,*

$$T_a(k, l, m) = i\sigma(\rho - \rho^{-1}, -\rho, \rho^{-1}), \quad \rho = \sqrt{a_l/a_k}, \quad \sigma = \frac{1}{2\sqrt{a_m}}. \quad (10)$$

In particular three-dimensional Euler, $a \equiv 1$, sits at $\frac{i}{2}(E_3 - E_2)$, with coefficients proportional to $(0, -1, 1)$.

Proof. The two terms of (3) contribute

$$a_k^{-1/2} a_l^{1/2} a_m^{-1/2} S(k, l, m)[a, b, c] \quad \text{and} \quad a_l^{-1/2} a_k^{1/2} a_m^{-1/2} S(l, k, m)[b, a, c].$$

By Lemma 4.2 and the substitutions above,

$$S(k, l, m)[a, b, c] = i(E_1 - E_2), \quad S(l, k, m)[b, a, c] = i(E_3 - E_1).$$

Halving the sum and collecting gives (10). $\square$

Degenerate triads here are the *collinear* ones, $k \parallel l$, on which all three E_j vanish identically; they are excluded throughout, as the $F = 0$ triangles were in Section 3.

5. RIGIDITY IN THREE DIMENSIONS

Theorem 5.1. *Write $T_a(k, l, m) = (\alpha, \beta, \gamma)$ in the basis (E_1, E_2, E_3) . Then*

$$-\beta/\gamma = a_l/a_k,$$

so the ratio of the symbol at two legs of a triad is determined pointwise by the vertex. Consequently the condition that a given vertex be Euler–Arnold is, in the unknown $\log a$, the linear system $\log a_l - \log a_k = \log(-\beta/\gamma)$ over all non-degenerate triads: a discrete gradient condition on the triad graph, whose solvability is a cocycle condition around cycles.

Proof. Immediate from (10): $-\beta/\gamma = \rho/\rho^{-1} = \rho^2$. $\square$

This is strictly stronger than the two-dimensional situation, where by Theorem 3.3 only *differences* of the symbol were readable, and only up to a triangle-dependent factor. The gain is a consequence of the per-triad vertex space having dimension two rather than one.

Theorem 5.2. *Let μ be a scalar triad reweighting of three-dimensional Euler, that is $\tilde{T} = \mu_\tau T_1$. Suppose the graph whose vertices are the wavevectors and whose edges are the pairs occurring in a common non-degenerate triad is connected. Then $\tilde{T}$ is Euler–Arnold if and only if μ is constant, and the corresponding symbols are the constants. The gauge is one-dimensional.*

Proof. By Proposition 4.3, T_1 has coefficient direction $(0, -1, 1)$ on every triad, and a scalar reweighting preserves that direction. If $\tilde{T} = T_{a'}$ then by (10) the direction $(\rho - \rho^{-1}, -\rho, \rho^{-1})$ with $\rho = \sqrt{a'_l/a'_k}$ must be proportional to $(0, -1, 1)$, which forces $\rho - \rho^{-1} = 0$, i.e. $\rho = 1$, i.e. $a'_k = a'_l$. Applying this to each of the three pairs of each non-degenerate triad, a' is constant along every edge of the graph, hence constant by connectedness. Then $\sigma = \frac{1}{2}(a')^{-1/2}$ is constant and the displayed formula forces μ constant. The converse is clear. $\square$

The connectedness hypothesis is a finite check for each truncation; on $\{|k| \leq K\} \subset \mathbb{Z}^3$ with $K = 2, 3, 4$ the graph has a single component.

Remark 5.3. Theorem 5.2 is the point at which two and three dimensions part company. In two dimensions the gauge was two-dimensional and its non-trivial direction was the α -model (Corollary 3.11); in three it is one-dimensional. This does not say that the three-dimensional α -model fails to be an Euler–Arnold system — it is one, with $a_k = 1 + \alpha^2|k|^2$ — but that it is no longer a *reweighting* of Euler. It points in a different direction of the two-dimensional per-triad space, which in two dimensions it could not do, that space being a line.

Remark 5.4 (Rigidity, and what does not follow). It is tempting to read Corollary 3.11 and Theorem 5.2 as a single argument: in two dimensions the one non-trivial deformation is the Camassa–Holm inertia operator, the viscous three-dimensional equation built from that operator is globally well-posed [10], and therefore the only deformation the structure admits is one that

does not blow up. That reading is wrong, and it is worth saying why, because the two statements are of quite different standing.

The two facts are these. In two dimensions the L^2 pairing is ad-invariant, so by (5) the tensor F is totally antisymmetric and the per-triad vertex space is a line: every Euler–Arnold vertex on a triangle is a scalar multiple of every other. Being Euler–Arnold and being a reweighting of Euler therefore coincide triangle by triangle, the entire content is global consistency, and Theorem 3.8 returns the two-parameter family of Corollary 3.11. In three dimensions the pairing is not ad-invariant (Lemma 4.1), the per-triad space is a plane, and the two notions come apart: $T_{a'}$ generally points in a different direction from T_1 , and Theorem 5.2 says the reweightings collapse to global rescaling.

They do not join, for the reason Theorem 5.2 itself supplies: the Camassa–Holm direction is *not* among the three-dimensional reweightings, so [10] is evidence about an object that Theorem 5.2 has placed outside the class it describes. The three-dimensional statement needs no input from [10] and is the stronger of the two; the two-dimensional one is context, and identifies which deformation the structure singles out when it admits one at all.

Two scope remarks. Theorem 5.2 concerns reweightings, and does not say that three-dimensional Euler–Arnold systems are rigid in general: the α -model is one, as Remark 5.3 records. And the rigidity is a statement about single deformations — Theorems 5.12 and 5.14 show that averages behave differently.

5.1. Multiplier-twisted vertices. Tao’s elementary operators have the form $T_1 B(T_2 u, T_3 v)$ with T_i mild. Taking the T_i to be scalar Fourier multipliers d_1, d_2, d_3 and writing the resulting trilinear form as $\tilde{T}$, we can characterise the Euler–Arnold members completely.

Theorem 5.5. *Let d_1, d_2, d_3 be positive even scalar symbols on $\mathbb{Z}^3 \setminus \{0\}$ and let $\tilde{T}$ be the corresponding twist of S , symmetrised in the first two slots.*

- (1) *$\tilde{T}$ conserves energy if and only if d_1 and d_3 are proportional.*
- (2) *Suppose $d_3 = c d_1$, and normalise $c = 1$ by absorbing the constant; write $d := d_1 = d_3$ and $e := d_2$. Then $\tilde{T}$ is Euler–Arnold if and only if $h := de$ is constant, in which case the symbol is $a = c'e/d$ and the twist is the pattern $(A^{-1/2}, A^{1/2}, A^{-1/2})$.*

Proof. (i) If $d_3 = c d_1$ then $\tilde{T}(X, X, X) = c S(d_1 X, d_2 X, d_1 X)$, whose first and third arguments agree, so it vanishes by Lemma 2.1. Conversely, on a triad $k + l + m = 0$ the cancellation condition reads

$$d_1(k)d_2(l)d_3(m) - d_1(l)d_2(k)d_3(m) - d_1(m)d_2(l)d_3(k) + d_1(m)d_2(k)d_3(l) = 0,$$

which factors as $d_2(l)W(k, m) = d_2(k)W(l, m)$, where

$$W(x, y) := d_1(x)d_3(y) - d_1(y)d_3(x)$$

is antisymmetric. The two cyclic images of the cancellation condition factor in the same way and give $d_2(m)W(k, l) = d_2(l)W(k, m)$ and $d_2(k)W(l, m) = -d_2(m)W(k, l)$. Chaining the three,

$$d_2(l)W(k, m) = d_2(k)W(l, m) = -d_2(m)W(k, l) = -d_2(l)W(k, m),$$

so $2d_2(l)W(k, m) = 0$ and hence $W(k, m) = 0$, the symbols being positive. Thus $d_1(k)/d_3(k) = d_1(m)/d_3(m)$ for every non-parallel pair, and since the triad graph is connected the ratio d_1/d_3 is constant.

(ii) On a triad $k + l + m = 0$ the two terms of the symmetrisation contribute, in the basis (E_1, E_2, E_3) , coefficients proportional to $(u - v, -u, v)$ where

$$u = d(k)e(l)d(m), \quad v = d(l)e(k)d(m).$$

Comparing with (10), $\tilde{T} = T_a$ requires $u = \sigma\rho$ and $v = \sigma\rho^{-1}$, hence

$$\sigma = \sqrt{uv} = d(m)\sqrt{h(k)h(l)}, \quad \rho^2 = u/v = \frac{d(k)e(l)}{d(l)e(k)} = \frac{g(k)}{g(l)}, \quad g := d/e.$$

Since $\sigma = 1/(2\sqrt{a_m})$ depends only on m , and $d(m)$ already does, the quantity $h(k)h(l)$ must depend only on $k + l = -m$. Writing $\psi = \log h$, this says $\psi(k) + \psi(l) = \Phi(k + l)$ for all admissible k, l . Hence $\psi(k) - \psi(k') = \psi(l') - \psi(l)$ whenever $k - k' = l' - l$, so $\psi(k) - \psi(k')$ depends only on $k - k'$ and defines an additive function on $\mathbb{Z}^3$, necessarily of the form $\lambda \cdot (k - k')$. Thus ψ is affine, $\psi(k) = c + \lambda \cdot k$; evenness of h forces $\lambda = 0$ and h is constant. Given that, $\rho^2 = g(k)/g(l) = a_l/a_k$ yields $a \propto 1/g = e/d$, and h constant makes $d \propto a^{-1/2}$, $e \propto a^{1/2}$. Conversely this pattern is Euler–Arnold by construction. $\square$

Remark 5.6. The direction of the vertex always fits: it forces $a \propto e/d$ regardless. It is the *scale* σ that carries the condition. Numerically, the fitted scale constant is constant to 10^{-4} when h is constant and varies by factors 9.2 and 50.0 in two test families where it is not.

5.2. Anisotropic inertia operators. Theorem 5.2 treats scalar symbols. We now allow A to be matrix-valued: A_k a symmetric positive operator on the plane $k^\perp$ with $A_{-k} = A_k$, which is three real parameters per $\pm$ pair rather than one. The conclusion does not improve.

Lemma 5.7 (Adapted frame). *Let $k + l + m = 0$ be non-degenerate, P the plane they span and ν a unit normal to P . Then ν lies in each of $k^\perp, l^\perp, m^\perp$, and with $\hat{k} := \nu \times k/|k|$, and $\hat{l}, \hat{m}$ defined likewise, the pairs $(\hat{k}, \nu)$, $(\hat{l}, \nu)$, $(\hat{m}, \nu)$ are orthonormal bases of $k^\perp, l^\perp, m^\perp$. Writing $a = a_1\hat{k} + a_2\nu$ and similarly for b, c , and putting $J := |k \times l|$, $x := |k|$, $y := |l|$, $z := |m|$,*

$$\begin{aligned} S(k, l, m)[a, b, c] &= i[\alpha a_1 b_1 c_1 + \beta a_2 b_2 c_1 + \gamma a_1 b_2 c_2], \\ S(l, k, m)[b, a, c] &= i[\alpha' a_1 b_1 c_1 + \beta' a_2 b_2 c_1 + \gamma' a_2 b_1 c_2], \end{aligned}$$

with

$$\alpha = \frac{Jy}{xz}, \quad \beta = -\frac{J}{z}, \quad \gamma = -\frac{J}{x}, \quad \alpha' = -\frac{Jx}{yz}, \quad \beta' = \frac{J}{z}, \quad \gamma' = \frac{J}{y}.$$

In particular every monomial occurring carries an even number of indices 2, so S , and hence T_a for scalar a , is even under the reflection $\nu \mapsto -\nu$.

Proof. Direct computation from (9), using $\hat{k} \cdot l = J/x$, $\hat{m} \cdot l = -J/z$, $\hat{m} \cdot k = J/z$, $\hat{l} \cdot k = -J/y$ and $\hat{k} \cdot \hat{l} = (k \cdot l)/(xy)$, together with $k \cdot l + l \cdot m = -|l|^2$ and $k \cdot l + k \cdot m = -|k|^2$. $\square$

Definition 5.8 (Transversality). A truncation is *transversal* if every k occurring in it lies in two non-degenerate triads whose plane normals ν_1, ν_2 satisfy $\nu_1 \cdot \nu_2 \notin \{0, \pm 1\}$.

Theorem 5.9. *Work in a transversal truncation whose triad graph is connected. Linearise at $A = I + \epsilon H$, with H_k symmetric on $k^\perp$ and $H_{-k} = H_k$. Then the set of H for which $\frac{d}{d\epsilon}|_0 T_{I+\epsilon H}$ is proportional to T_1 on every non-degenerate triad is exactly the one-dimensional space spanned by $H_k = I|_{k^\perp}$.*

Proof. Since $A_k^{\pm 1/2} = I \pm \frac{\epsilon}{2} H_k + O(\epsilon^2)$, differentiating (3) gives

$$\begin{aligned} \dot{T}[H] &= \frac{1}{4} \left\{ -S(H_k a, b, c) + S(a, H_l b, c) - S(a, b, H_m c) \right. \\ &\quad \left. - S'(H_l b, a, c) + S'(b, H_k a, c) - S'(b, a, H_m c) \right\}, \end{aligned}$$

where $S' := S(l, k, m)$. Fix a non-degenerate triad and use the frame of Lemma 5.7, writing $H_k = p_k \hat{k} \otimes \hat{k} + q_k (\hat{k} \otimes \nu + \nu \otimes \hat{k}) + r_k \nu \otimes \nu$, and similarly at l and m . The entries p, r preserve the parity of Lemma 5.7 while the entries q reverse it, so $\dot{T}[H]$ splits into an even part depending

on (p, r) and an odd part depending linearly on (q_k, q_l, q_m) . As T_1 is even, the hypothesis forces the odd part to vanish.

Collecting the coefficient of $a_2 b_2 c_2$ in the odd part gives $-q_k \gamma - q_l \gamma' - q_m(\beta + \beta') = 0$. By Lemma 5.7, $\beta + \beta' = 0$, $\gamma = -J/x$ and $\gamma' = J/y$, so this reads $J(q_k/x - q_l/y) = 0$, that is $q_k/|k| = q_l/|l|$. The two cyclic images of the triad give the same relation for the other pairs, so

$$\frac{q_k}{|k|} = \frac{q_l}{|l|} = \frac{q_m}{|m|} =: t.$$

Collecting the coefficient of $a_2 b_1 c_1$ gives $q_k(\alpha' - \alpha) + q_l(\beta - \beta') - q_m \gamma' = 0$; substituting the values of Lemma 5.7 and clearing $-J/(xyz)$ this is

$$q_k(x^2 + y^2) + 2q_l xy + q_m xz = 0.$$

Putting $(q_k, q_l, q_m) = t(x, y, z)$ turns the left side into $t x (x^2 + 3y^2 + z^2)$, which is positive unless $t = 0$. Hence $t = 0$ and $q_k = q_l = q_m = 0$.

So H_k is diagonal in the adapted basis $(\hat{k}, \nu_\tau)$ of every non-degenerate triad τ containing k . A symmetric operator on a plane with distinct eigenvalues has a unique orthonormal eigenbasis up to order and sign; by transversality k lies in two triads whose adapted bases have distinct axis pairs, so the eigenvalues coincide and $H_k = h_k I|_{k^\perp}$.

Finally, for $H_k = h_k I$ the operator $A = I + \epsilon H$ is the scalar symbol $a = 1 + \epsilon h$, and differentiating (10) at $\epsilon = 0$ gives coefficients

$$i\left(\frac{1}{2}(h_l - h_k), \frac{1}{4}h_m - \frac{1}{4}(h_l - h_k), -\frac{1}{4}h_m - \frac{1}{4}(h_l - h_k)\right),$$

which is proportional to $T_1 \propto (0, -1, 1)$ precisely when $h_k = h_l$. Applying this on every non-degenerate triad and using connectedness of the triad graph, h is constant. Conversely $H_k = I|_{k^\perp}$ is the derivative of a global rescaling and does satisfy the condition. $\square$

Remark 5.10. Transversality and connectedness are finite conditions on the truncation alone. Both hold for $\{|k| \leq K\} \subset \mathbb{Z}^3$ with $K = 2, 3, 4$; at $K = 3$ all 122 modes lie in two triads with $\nu_1 \cdot \nu_2 \notin \{0, \pm 1\}$. The conclusion was also confirmed directly: the kernel has dimension one for $\{|k| \leq 2\}$ (48 unknowns, 544 conditions, rank 47) and $\{|k| \leq 3\}$ (183, 8736, rank 182), and likewise at three random matrix-valued base points, with a singular-value gap of ten to eleven orders.

5.3. Combinations. Since S is rotation invariant, conjugating by a rotation acts on the symbols alone, $d \mapsto d \circ R$, so a rotation-averaged operator is a linear combination of multiplier twists. Theorem 5.5 settles a single twist. For combinations, the energy condition is again explicit.

Proposition 5.11. *Let $\tilde{T}_j$ be the twist by (d_1^j, d_2^j, d_3^j) , put $W_j(x, y) := d_1^j(x)d_3^j(y) - d_1^j(y)d_3^j(x)$, and let $c_j \in \mathbb{R}$. Then $\sum_j c_j \tilde{T}_j$ conserves energy if and only if*

$$\sum_j c_j d_2^j(m) W_j(k, l) = 0 \tag{11}$$

for every non-degenerate triad $k + l + m = 0$. Equivalently, writing $\phi_j := d_1^j/d_3^j$,

$$\sum_j c_j d_2^j(m) d_3^j(k) d_3^j(l) [\phi_j(k) - \phi_j(l)] = 0.$$

Proof. As in the proof of Theorem 5.5(i), the three components of the cancellation defect of a single twist on the triad are $P_j - Q_j$, $-P_j + R_j$ and $Q_j + R_j$, where $P_j := d_2^j(l)W_j(k, m)$, $Q_j := d_2^j(k)W_j(l, m)$ and $R_j := d_2^j(m)W_j(k, l)$. The defect is linear in the vertex, so for the combination the three components are the same expressions with P, Q, R replaced by

$\sum_j c_j P_j$ and so on. Setting all three to zero gives $\sum c_j P_j = \sum c_j Q_j$ and $\sum c_j P_j = \sum c_j R_j$ and $\sum c_j Q_j = -\sum c_j R_j$; the first two give $\sum c_j Q_j = \sum c_j R_j$, which with the third forces $\sum c_j R_j = 0$ and hence all three vanish. The converse is immediate. $\square$

Condition (11) does admit non-trivial solutions, with positive weights: rotation averaging can manufacture energy conservation out of terms none of which conserves it.

Theorem 5.12. *For $v \in S^2$ and $\epsilon > 0$ let $\tilde{T}_v$ be the twist with*

$$d_1^{(v)}(x) := 1 + \epsilon(v \cdot \hat{x})^2, \quad d_2 := 1, \quad d_3 := 1, \quad \hat{x} := x/|x|.$$

These are rotation averages of a single twist: $d_1^{(v)} = d_1^{(u)} \circ R$ for any rotation with $R^\top u = v$, and d_2, d_3 are rotation invariant. Then

- (1) *no single $\tilde{T}_v$ conserves energy;*
- (2) *for $c_j > 0$ and $v_j \in S^2$, the average $\sum_j c_j \tilde{T}_{v_j}$ conserves energy if and only if*

$$\sum_j c_j v_j v_j^\top \propto I,$$

a tight-frame condition, satisfied for instance by an orthonormal triple with equal weights.

Proof. (i) $d_1^{(v)}$ is non-constant while $d_3 = 1$, so $d_1^{(v)}$ is not proportional to d_3 and Theorem 5.5(i) applies.

(ii) Here $W_j(x, y) = d_1^{(v_j)}(x) - d_1^{(v_j)}(y)$ and $d_2^j \equiv 1$, so (11) reads

$$\sum_j c_j [d_1^{(v_j)}(k) - d_1^{(v_j)}(l)] = \epsilon [\hat{k}^\top A \hat{k} - \hat{l}^\top A \hat{l}] = 0, \quad A := \sum_j c_j v_j v_j^\top,$$

for every non-degenerate triad. Since k and l range over pairs of non-parallel directions, this says that $\hat{x}^\top A \hat{x}$ is constant on the sphere, which for symmetric A holds precisely when $A \propto I$. Proposition 5.11 then gives the claim. $\square$

Remark 5.13. So the characterisation of Theorem 5.5 does not survive averaging. Numerically, each term of the orthonormal-triple example has cancellation defect 0.4 while the equally weighted sum has defect 7×10^{-16} ; a non-tight triple gives 0.8. This is precisely the kind of freedom that independent per-leg rotations are introduced for in (12), and together with Theorem 5.14 it shows that neither the energy condition nor the Euler–Arnold property can be checked term by term once averaging is allowed.

For the Euler–Arnold property itself, averaging generically leaves the class: over 25 random rotation averages at $\{|k| \leq 3\}$ the relative residual of the linear system of Theorem 5.1 has median 0.29 to 0.46 for 2 to 8 terms, against 4×10^{-15} for a single Euler–Arnold twist. But the class is not simply unstable under averaging: there is an explicit non-trivial family on which it is closed, which we record because it limits what can be expected of this route.

Theorem 5.14. *Let $\epsilon \in \mathbb{Z}^3$ and let $\pi(k) := \epsilon \cdot k \bmod 2 \in \{0, 1\}$. Let b be a positive even symbol, let $c > 0$, and put $a^1 := b e^{2c\pi}$ and $a^2 := b$, so that a^1/a^2 is non-constant. Then for all $c_1, c_2 > 0$,*

$$c_1 T_{a^1} + c_2 T_{a^2} = T_a, \quad a = \frac{b \kappa^{-\pi}}{(c_1 + c_2)^2}, \quad \kappa := \frac{c_1 e^{-2c} + c_2}{c_1 + c_2}.$$

In particular an average of Euler–Arnold vertices with distinct symbols can again be Euler–Arnold.

Proof. Write $\chi := c\pi$, so $a^1 = b e^{2\chi}$. With ρ_j, σ_j as in (10) for the symbol a^j , we have $\sigma_1 = \sigma_2 e^{-\chi(m)}$ and $\rho_1 = \rho_2 e^{\chi(l) - \chi(k)}$, so on a triad

$$U = \sigma_2 \rho_2 [c_1 e^\theta + c_2], \quad V = \sigma_2 \rho_2^{-1} [c_1 e^{\theta'} + c_2],$$

where $\theta := -\chi(m) + \chi(l) - \chi(k)$ and $\theta' := -\chi(m) + \chi(k) - \chi(l)$. Note $\theta' \neq -\theta$ in general.

Since $k + l + m = 0$ we have $\pi(k) + \pi(l) + \pi(m) \equiv 0 \pmod{2}$, so $(\pi(k), \pi(l), \pi(m))$ is one of $(0, 0, 0)$, $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$, and correspondingly (θ, θ') is $(0, 0)$, $(0, 0)$, $(-2c, 0)$, $(0, -2c)$. Hence

$$\frac{U}{V} = \rho_2^2 \kappa^{\pi(k) - \pi(l)}, \quad UV = \sigma_2^2 (c_1 + c_2)^2 \kappa^{\pi(m)},$$

in each of the four cases. The second depends only on m , and equals $1/(4a_m)$ precisely for $a = b \kappa^{-\pi}/(c_1 + c_2)^2$; the first then equals a_l/a_k for that same a , since $\rho_2^2 = b_l/b_k$. By (10) these two identities say exactly that the combination is T_a . $\square$

Remark 5.15. Theorem 5.14 is the reason the second half of this subsection is stated as a generic rather than a universal claim. The parity functions are precisely the non-constant χ for which $\chi(l) - \chi(k)$ is controlled by $k + l$ alone, which is what the scale condition needs; a generic χ does not have that property, and the average then leaves the class.

6. TAO'S AVERAGED OPERATOR

6.1. The operator. Tao's admissible class [19, (1.12)] is

$$\langle \tilde{B}(u, v), w \rangle := \mathbb{E} \langle B(m_1(D) \text{Rot}_{R_1} \text{Dil}_{\lambda_1} u, m_2(D) \text{Rot}_{R_2} \text{Dil}_{\lambda_2} v), m_3(D) \text{Rot}_{R_3} \text{Dil}_{\lambda_3} w \rangle, \quad (12)$$

with m_i real order-zero Fourier multipliers, R_i rotations and λ_i dilations, all random and independent across the three legs. Theorem 5.5 settles the sub-class in which the rotations and dilations agree across legs, since by [19, Remark 1.4] one of each may then be removed.

The operator used for blowup is a *local cascade operator* [19, Def. 3.1, (4.1)]. Fix m profiles $\psi_1, \dots, \psi_m \in H_{df}^{10}(\mathbb{R}^3)$ with Fourier transforms supported on disjoint balls $B_i \cup -B_i$ inside the single thin annulus $1 < |\xi| \leq 1 + \varepsilon_0/2$, and set $\psi_{i,n}(x) := (1 + \varepsilon_0)^{3n/2} \psi_i((1 + \varepsilon_0)^n x)$. Then

$$C(u, v) := \sum_{n \in \mathbb{Z}} \sum_{(i_1, i_2, i_3, \mu) \in \{1, \dots, m\}^3 \times \mathcal{S}} \alpha_{i_1 i_2 i_3 \mu} (1 + \varepsilon_0)^{5n/2} \langle u, \psi_{i_1, n+\mu_1} \rangle \langle v, \psi_{i_2, n+\mu_2} \rangle \psi_{i_3, n+\mu_3}, \quad (13)$$

with $\mathcal{S} = \{(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$, subject to a symmetry condition [19, (4.2)] and a cancellation condition [19, (4.3)]. In [19, §6.1] one takes $m = 4$ and the sixteen non-zero structure constants of Table 1 there.

Remark 6.1. We reconstructed that table and verified that it satisfies [19, (4.2)] exactly and [19, (4.3)] to 1.8×10^{-15} . Two entries have their exponentials damaged in the electronic text, and we record carefully how they are fixed, since Corollary 6.4 is stated against this reconstruction. The two conditions [19, (4.2)–(4.3)] do *not* settle them: both candidate readings, $\varepsilon^2 e^{-K^{10}}$ and $\varepsilon^2 e^{-K^{-10}}$, satisfy symmetry and cancellation identically. They are settled instead by the equations of motion [19, (6.3)–(6.6)], which Tao also prints and which are an independent consequence of the table through [19, (4.10)]: expanding our reconstruction through [19, (4.10)] reproduces [19, (6.3)–(6.6)] term by term for the reading $\varepsilon^2 e^{-K^{10}}$ and fails for the other. We adopt that reading.

Corollary 6.4 is in any case insensitive to this. Theorem 6.3 depends only on which (i_1, i_2, i_3, μ) carry a non-zero structure constant, not on the values, and the two readings have identical support; the count of twelve is the same for either.

Integrating the reconstructed equations of motion from the stated initial data reproduces the cascade, energy moving up eight scales with the total non-increasing. The asymptotic regime $1 \ll \varepsilon_0^{-1} \ll K \ll \varepsilon^{-1} \ll n_0$ needed for the blowup itself is not numerically accessible.

6.2. A range obstruction.

Theorem 6.2. *Let C be any operator of the form (13), with any structure constants. Then C is not of the form B_A for any symmetric positive definite A .*

Proof. By (13), $C(u, v)$ is a combination of the $\psi_{i_3, \cdot}$, so the range of C lies in $W := \overline{\text{span}}\{\psi_{i,n}\}$, which has m directions per scale. Put $P := A^{-1/2}$, symmetric positive and in particular invertible, so that $B_A(u, v) = P \text{ad}_{P^*u}^*(P^{-1}v)$ up to the symmetrisation. If $B_A = C$ then $\text{range}(B_A) \subseteq W$; since P is invertible, Pu and $P^{-1}v$ range over all of $\mathfrak{g}$ as u, v do, so this forces $\text{ad}_x^*(y) \in P^{-1}W$ for all $x, y \in \mathfrak{g}$. Thus the whole range of ad^* would lie in a single closed subspace with m directions per scale. It does not: on the truncation $\{|k| \leq 2\}$ of $\mathbb{T}^3$, of real dimension 64, the span of $\{\text{ad}_u^* u\}$ is all of it, whereas m directions per frequency shell supply 16; on $\mathbb{R}^3$ each annulus carries a continuum of divergence-free directions and (13) uses m of them. $\square$

Theorem 6.2 is valid but blunt: it would equally exclude any operator of finite rank per scale, and says nothing about the Lie structure. The next result is the substantive one.

6.3. A geometric obstruction.

Theorem 6.3. *Let A be a Fourier multiplier, so that T_A is supported on $\{\xi_1 + \xi_2 + \xi_3 = 0\}$. Suppose the balls B_i have radius small compared with ε_0 , and $0 < \varepsilon_0 < 1$. Then T_A vanishes identically on every mode triple of (13) carrying fewer than three distinct species, that is, whenever $|\{i_1, i_2, i_3\}| \leq 2$.*

Proof. The three legs carry frequencies $\xi_j \in \pm(1 + \varepsilon_0)^{n+\mu_j}(b_{i_j} + O(r))$, where b_i is the centre of B_i , r its radius, and the sign accounts for the support being $B_i \cup -B_i$. A non-zero contribution to T_A requires $\xi_1 + \xi_2 + \xi_3 = 0$ for some admissible choice, hence, to leading order in r ,

$$\sum_{j=1}^3 \sigma_j (1 + \varepsilon_0)^{\mu_j} b_{i_j} = 0, \quad \sigma_j \in \{\pm 1\}.$$

Legs carrying the same species carry the same vector b_i , so grouping by species this reads $\sum_i c_i b_i = 0$ with $c_i = \sum_{j: i_j=i} \sigma_j (1 + \varepsilon_0)^{\mu_j}$.

Suppose exactly two species occur, say i twice and i' once, and let μ be the offset of the single leg. Then $c_{i'} = \pm(1 + \varepsilon_0)^\mu \neq 0$. If $c_i = 0$ the relation reduces to $c_{i'} b_{i'} = 0$, impossible since $|b_{i'}| > 1$. Otherwise $c_i b_i + c_{i'} b_{i'} = 0$ forces $b_i \parallel b_{i'}$ and $|b_i|/|b_{i'}| = |c_{i'}|/|c_i|$. With $\mu_j \in \{0, 1\}$ the possible values of $|c_i|$ for a doubled species are 0, 2, $1 + (1 + \varepsilon_0)$ and ε_0 , while $|c_{i'}| \in \{1, 1 + \varepsilon_0\}$; in every case the required ratio $|c_{i'}|/|c_i|$ lies outside the interval $[(1 + \varepsilon_0/2)^{-1}, 1 + \varepsilon_0/2]$ to which $|b_i|/|b_{i'}|$ is confined by the annulus, once $\varepsilon_0 < 1$. If only one species occurs, $c_i = \pm 1 \pm 1 \pm 1 \neq 0$ when all μ_j agree, and the relation reduces to $c_i b_i = 0$, again impossible. Shrinking r absorbs the $O(r)$ error. $\square$

Corollary 6.4. *With $m = 4$ and the structure constants of [19, Table 1], twelve of the sixteen non-zero $\alpha_{i_1 i_2 i_3 \mu}$ carry fewer than three distinct species. On the corresponding mode triples T_A vanishes identically for every Fourier-multiplier A , while $\alpha \neq 0$. Hence $C \neq B_A$ for any such A .*

The four exceptions are $(3, 4, 1)$, $(4, 3, 1)$, $(1, 3, 4)$ and $(3, 1, 4)$, all with $\mu = 0$ and all carrying three distinct species: the ε^{-2} gate. The twelve that fail include the terms driving the cascade —

the $X_1^2 \rightarrow X_2$ pump of size ε , the $X_3^2 \rightarrow X_2$ term of size K^{10}/ε , and the $\varepsilon^2 e^{-K^{10}}$ seed. They also include every term that couples one scale to the next, which sharpens Corollary 6.4 considerably.

Corollary 6.5. *Exactly three of the sixteen non-zero structure constants carry an offset $\mu \neq (0, 0, 0)$: the term $\alpha_{4,4,1,(0,0,1)}$ and its two partners $\alpha_{1,4,4,(1,0,0)}$, $\alpha_{4,1,4,(0,1,0)}$. All three carry fewer than three distinct species, hence all three lie among the twelve of Corollary 6.4; and the four couplings not excluded all have $\mu = (0, 0, 0)$. Consequently every coupling of (13) that transfers energy between scales is excluded, and the part left untouched by Theorem 6.3 is single-scale. Explicitly, setting the excluded twelve to zero in [19, (4.10)] leaves, at each scale n separately and modulo dissipation,*

$$\partial_t X_{1,n} = -\varepsilon^{-2}(1 + \varepsilon_0)^{5n/2} X_{3,n} X_{4,n}, \quad \partial_t X_{4,n} = \varepsilon^{-2}(1 + \varepsilon_0)^{5n/2} X_{1,n} X_{3,n},$$

with $\partial_t X_{2,n} = \partial_t X_{3,n} = 0$. This conserves $X_{1,n}^2 + X_{4,n}^2$ at every scale, so the surviving sub-operator is globally bounded and cannot cascade.

Proof. The first two assertions are read off [19, Table 1] and Theorem 6.3. For the displayed system, the four surviving constants are $\alpha_{3,4,1} = \alpha_{4,3,1} = -\varepsilon^{-2}/2$ and $\alpha_{1,3,4} = \alpha_{3,1,4} = \varepsilon^{-2}/2$, all at $\mu = 0$; substituting into [19, (4.10)] gives the two equations, and no term has $i_3 \in \{2, 3\}$, so $X_{2,n}$ and $X_{3,n}$ are constant. Then $\frac{d}{dt}(X_{1,n}^2 + X_{4,n}^2) = 2\varepsilon^{-2}(1 + \varepsilon_0)^{5n/2}(-X_{1,n}X_{3,n}X_{4,n} + X_{4,n}X_{1,n}X_{3,n}) = 0$. $\square$

So the split is sharp in a way Corollary 6.4 alone does not convey: of Tao’s sixteen couplings, the Euler bracket can accommodate only ones that act within a single scale, and the entire mechanism by which his construction moves energy to finer scales lies in the part it cannot accommodate.

Remark 6.6. This is what the independent per-leg rotations and dilations in (12) are for. Tao’s Theorem 3.2 shows that these otherwise inadmissible triples can be realised as averages of B precisely by rotating and dilating each leg separately; a footnote records that dilation averaging had to be added because a non-degeneracy condition failed without it. In the terminology of this paper, the averaging manufactures couplings that the Euler bracket does not possess.

Remark 6.7 (On the two obstructions). Theorems 6.2 and 6.3 are of different characters, and it is worth saying which does what. Theorem 6.2 holds for every symmetric positive definite A but is blunt: it would equally exclude any operator of finite rank per scale, and uses nothing about the Lie structure. Theorem 6.3 uses the structure and is quantitative, but needs A to be a Fourier multiplier.

That hypothesis is a normalisation rather than a technical convenience. Translations of $\mathbb{T}^3$ are themselves volume-preserving diffeomorphisms, and A is a Fourier multiplier precisely when the right-invariant metric it defines is in addition invariant under that translation subgroup. Every inertia operator arising from a geometric construction has this property, including $A = \text{id}$ for Euler and $A = I - \alpha^2 \Delta$ for the α -model. What is genuinely left open is a structural, quantitative obstruction for inertia operators that are not translation covariant; for those, only Theorem 6.2 applies.

7. SCOPE

The sharpest limitation is one of quantifier, and it is now possible to say where it bites. What Section 6 shows is that *Tao’s* averaged operator is not Euler–Arnold. It does not show that no Euler–Arnold operator blows up, and Theorem 5.14 was a reason to be careful about the stronger reading: the class is closed under some non-trivial averages. The caution was warranted, because the stronger reading is false.

In [20] Tao writes the Euler equation in vorticity–vector potential form $\partial_t \omega + \mathcal{L}_u \omega = 0$, $u = \delta A \omega$, and replaces the vector potential operator by a general operator A of order -2 . The Lie transport is untouched, so the bracket is fixed exactly as it is fixed here, and when A is positive definite the resulting equation is (2): it is the geodesic flow on $\text{SDiff}(M)$ for the right-invariant metric of quadratic form $\int \langle \omega, A \omega \rangle$ [20, Rem. 3.1], his A being the inverse of ours up to curl, and $A = \text{id}$ being the true Euler case. Energy, the Kelvin circulation theorem, helicity and vorticity transport all survive the replacement. His Theorem 1.13 then exhibits, on $\mathbb{R}^3$, a formally self-adjoint A with

$$(1 - \varepsilon) \|\omega\|_{H^{-1}}^2 \leq \int \langle \omega, A \omega \rangle \leq (1 + \varepsilon) \|\omega\|_{H^{-1}}^2$$

and smooth compactly supported initial vorticity admitting no solution on the unit time interval. So an Euler–Arnold operator does blow up, and the metric witnessing it is within ε of the Euler metric itself. Energy identity, Lie–Poisson form and vorticity transport are jointly insufficient for regularity, in the same sense in which [19] made energy identity and scaling jointly insufficient.

What this leaves open is not nothing, and it is the case the present paper is about. Tao’s operators are “inhomogeneous and non-isotropic” [20, Rem. 1.19] — not translation invariant, with conservation of impulse and of moment of impulse lost accordingly — and his kernel bound [20, Def. 1.7] fixes the order of A at -2 , the order of the Euler operator itself. The construction therefore holds the criticality of the metric fixed and breaks its homogeneity; it does not vary the smoothing power of A at all. The same remark records that translation invariance, rotation invariance and dilation invariance together return the class to true Euler up to normalising constants. Keeping translation and rotation invariance and dropping only dilation invariance leaves the isotropic multipliers, the operators acting on each divergence-free plane by a scalar symbol $a(|k|)$. That family is not reduced to a point, and it is where the question is sharpest. Hence:

Question 7.1. *Is there a positive definite isotropic Fourier multiplier A , with symbol $a(|k|)$, whose Euler–Arnold equation (2) on $\text{SDiff}(\mathbb{T}^3)$ or $\text{SDiff}(\mathbb{R}^3)$ blows up in finite time from smooth data?*

The anisotropic translation-covariant multipliers lie between this class and Tao’s inhomogeneous operators, and the question is open there too. We pose the isotropic form because it is the one whose one-dimensional counterpart is understood, and because a positive answer in the smallest class is the strongest refutation. The restriction is not one the methods here need: Theorem 6.3 is stated for translation-covariant A , and Theorem 5.9 already admits anisotropic, matrix-valued inertia operators.

Two remarks on how to read it. First, it is not a route to regularity. A negative answer contains $A = \text{id}$ and would prove global regularity for three-dimensional Euler as a corollary, so that direction is strictly harder than the problem it was meant to inform. A positive answer is a construction, and it would refute the programme of this paper outright. The asymmetry is the argument for posing the question early rather than late.

Second, the one-dimensional family is less of a reassurance than it looks, and the comparison runs in an unfavourable direction in three separate ways. The analogue on $\text{Diff}(S^1)$ of the symbols $a(|k|)$ is the family of right-invariant H^s metrics, $a(|k|) = |k|^{2s}$. There the geodesic flow is complete for $s > 3/2$ [9] and at the critical index $s = 3/2$ [3]. On the other side, however, blowup is known only at isolated orders — $s = 0$, which is Burgers, $s = 1/2$, which is the modified Constantin–Lax–Majda equation, and $s = 1$, which is Camassa–Holm — and that it occurs for every $s < 3/2$ is a conjecture of [3] rather than a theorem. Nobody has

interpolated between those three points, in either direction, in the one setting where both ends are understood. There is no monotonicity principle here to import.

Nor is the order a complete invariant. Camassa–Holm and Hunter–Saxton have the same highest-order term, yet all solutions of the second break down in finite time while some solutions of the first stay smooth forever [3]: the lower-order terms decide it. The one transfer result available runs at a fixed order rather than across orders, and only at the critical one — [3] shows that if all solutions remain smooth for a single metric of order $3/2$, the same holds for every metric of that leading order. So even in one dimension there is no well-defined answer attached to s for a continuity argument in s to propagate.

In three dimensions the smoothing end of the family is anchored, though not where one first looks for it. The inviscid Euler– α system is the case $s = 1$ and its global well-posedness in three dimensions is open, but that places the threshold above 1 rather than leaving the end free. For $s > \frac{1}{2} \dim M + 1$, which here reads $s > 5/2$, the geodesic flow of a strong right-invariant H^s metric is geodesically and metrically complete, on the subgroup of volume-preserving diffeomorphisms as well as on the full group [6, Rems. 7.3 and 7.5, Cor. 7.6]. The mechanism deserves naming, because it bounds what the geometry can be credited with: above $\frac{1}{2} \dim M + 1$ the conserved energy $\|u\|_{H^s}^2$ controls $\|\nabla u\|_{L^\infty}$ by Sobolev embedding alone, so no Beale–Kato–Majda quantity can diverge. Neither the bracket nor incompressibility is used; volume preservation enters only as the observation that SDiff is a closed subgroup. The same count in one dimension returns $3/2$, which is precisely the threshold of [9] and [3]. Nor does the mechanism stop exactly at the threshold. At $s = \frac{1}{2} \dim M + 1$ itself the energy misses L^∞ control of ∇u by a logarithm, which a Brezis–Gallouët argument pays for, so the endpoint is complete as well [12]. Pushed through Osgood’s lemma, the same argument reaches a whole logarithm below it. It covers every inertia symbol $|k|^5/g(|k|)^2$ with $\int_0^\infty (\int_0^T g(e^\tau)^2 d\tau)^{-1/2} dT = \infty$, among them $|k|^5/\log(e + |k|)$, again without the bracket or incompressibility, and verbatim on the full group [18, Prop. 7.4 and Rem. 7.5].

The direction one would want to travel from the other end is meanwhile unfavourable twice over. Raising s adds smoothing, which is the operation used to destroy blowup rather than to propagate it. More decisively, the interval ends at Euler: the position of the threshold relative to $s = 0$ is the question of finite-time blowup for three-dimensional Euler, so any principle carrying blowup from some $s_0 < 0$ upward to $s = 0$ would settle that question as a corollary. One should expect to reach blowup only at orders bounded away from zero, with the gap to zero carrying the whole of the difficulty.

Two recent results bear on this directly, and they cut both ways. Bauer, Preston and Valletta prove finite-time breakdown for higher-order EPDiff equations in every dimension $n \geq 3$ [4], and then, for the homogeneous inertia operator $A = (-\Delta)^k$, at every integer order $0 \leq k < \frac{1}{2}n + 1$, for radial data satisfying a sign condition on the initial momentum [5]. That is precisely the family $a(|k|) = |k|^{2s}$ of Question 7.1, with the same homogeneity; only the group differs. Taken with the completeness threshold above, it says that on the full diffeomorphism group in three dimensions the Sobolev value is *attained* at integer orders — breakdown at $k = 0, 1, 2$ and completeness above $5/2$ — so that there the Euler–Arnold structure buys nothing whatever, and only the non-integer window $(2, 5/2)$ is left open. But their equations are geodesic flows on that full group rather than on SDiff , and their criterion is that the Jacobian of the Lagrangian flow map reaches zero in finite time. On SDiff that Jacobian is identically one, so the criterion is vacuous on the class of Question 7.1. The two constructions nearest to that question each surrender exactly what it holds fixed: [20] keeps the volume-preserving bracket and breaks homogeneity instead of varying the order, while [4, 5] vary the order but require a group in

which volume can concentrate. A positive answer to Question 7.1 would therefore have to be constructed outright, by a mechanism compatible with incompressibility.

There is, on the other hand, a quantity to locate on SDiff itself, and pushing it downward is the constructive companion to Question 7.1. Neither monotonicity in s nor connectedness of the completeness region is available, for the reasons just given, so we define it in a way that assumes neither. Write $\mathcal{C}$ for the set of orders s at which (2) with $a(|k|) = |k|^{2s}$ is geodesically complete on $\text{SDiff}(\mathbb{T}^3)$, and put

$$s_c := \inf \{ \sigma \in \mathbb{R} : (\sigma, \infty) \subseteq \mathcal{C} \}. \quad (14)$$

The set on the right is upward closed by construction, so s_c is well defined whatever the shape of $\mathcal{C}$ — in particular without assuming that $\mathcal{C}$ is an interval — and $s_c \leq 5/2$ is immediate from [6].

Question 7.2. *Is $s_c < 5/2$? Equivalently, is the geodesic flow complete for some isotropic translation-covariant metric of order strictly below the Sobolev threshold $\frac{1}{2} \dim M + 1$?*

This is the thesis of the present paper in quantitative form. Above $\frac{1}{2} \dim M + 1$ completeness is bought by Sobolev embedding and the Euler–Arnold structure does no work at all. The same holds at that value, and within a logarithm below it, by the Brezis–Gallouët–Osgood argument above. So the region bought by embedding is a logarithmic neighbourhood of the Sobolev value, not the value itself. Anything gained beyond that neighbourhood, and in particular any gain of a power, has to be paid for with the structure itself. The full diffeomorphism group serves as a control, and it is a sharp one: by [5] the Sobolev value is attained there at every integer order, so on the full group the structure buys nothing at all. On SDiff nothing is known at any order $s < 5/2$ in either direction; what is known below the Sobolev value lies within a logarithm of it and is bought without the structure. Whether incompressibility moves s_c appreciably beneath the embedding value — and whether it moves it past 0, which is once more the global regularity of three-dimensional Euler — is open, and a theorem placing $s_c \leq 5/2 - \delta$ for any explicit $\delta > 0$ would be the first quantitative evidence that the structure does anything. Because the control supplies a baseline of zero, any $\delta > 0$ is attributable: it would measure what volume preservation is worth, and not what the Euler–Arnold form is worth, the latter having been shown on the full group to be worth nothing.

Two further facts from [18] bear on the expectation, and they pull opposite ways. First, $5/2$ is not an artefact of the estimates but the scaling-critical exponent: the conserved energy is invariant under the dilations preserving (2) exactly at $s = \frac{1}{2} \dim M + 1$, which is $5/2$ here and $3/2$ in one dimension, the latter being the sharp threshold on $\text{Diff}(S^1)$. Second, in two dimensions $\text{curl } m$ is a scalar, the stretching term is absent outright, and the flow on $\text{SDiff}(\mathbb{T}^2)$ is globally regular for *every* $s > 0$ — the entire supercritical range, s^* being 2 there. Volume preservation therefore buys nothing on the full group and everything on the volume-preserving group one dimension down, and neither analogy determines three dimensions. What distinguishes it is that the stretching term neither vanishes, as in $d = 2$, nor acts unimpeded, as on Diff .

One prerequisite deserves recording, because Question 7.2 cannot be approached without it and it has since been supplied. Displacing s_c below the embedding threshold requires a continuation criterion — a statement that the solution persists for as long as some quantity remains integrable in time — and none was available for the family $a(|k|) = |k|^{2s}$ on SDiff . Tao records the gap and leaves it open: analogues of at least some of the known refinements should be establishable, the generalised Biot–Savart law obeying estimates very similar to the true one, but he does not attempt it [20, Rem. 1.10]. The companion note [18] supplies one. For $s > 0$ and $r > \frac{5}{2}$ the solution persists as long as $\int_0^T \|\text{curl } m\|_{L^p} dt < \infty$ for a single exponent $p > 3/(2s)$, and the classical Beale–Kato–Majda criterion is the endpoint $s \rightarrow 0^+$ of that family

rather than an instance of it. It is needed whether the answer to Question 7.2 is positive or negative, and it settles neither.

One route deserves closing off explicitly, because it suggests itself and it does not work. If the Euler–Arnold structure is to be spent on Question 7.2, the natural currency is the conserved quantities, and beyond the energy three-dimensional ideal flow carries one: the helicity of the momentum, a Casimir of the Lie–Poisson structure and hence conserved for every inertia operator. The companion paper [17] reduces it, for translation-covariant A , to the numbers $\sqrt{\det A_k}$, one for each pair $\{k, -k\}$, where A_k is the restriction of A to the plane $k^\perp$; and the reduction applies to the value of the helicity and not only to its conservation, so the entire Casimir content of such an A is that one positive vector. Two consequences, pointing the same way. On the isotropic family of Question 7.1 the map is injective, $\det A_k = a(|k|)^2$, so the helicity carries no information about A beyond A itself; being indefinite it controls no norm, and it cannot enter a continuation argument at all. On the anisotropic translation-covariant multipliers, where the question is open too, the map is far from injective: A_k is a positive definite form on a two-plane and carries three parameters, of which the determinant pins one. Along the two that remain at each mode — the ratio of the eigenvalues and the orientation of the principal axes inside $k^\perp$ — the helicity is constant to machine precision, while the vortex-stretching production whose integrability [18] requires varies by a factor of several [17, Prop. 6.4]. The energy does move along that fibre; but the energy is exactly the quantity already known to decide nothing below $5/2$. Either way the invariants fail to reach the quantity that settles the question, and a displacement of s_c will have to be bought with an estimate rather than found as a conservation law.

A second route suggests itself, and others have followed it to its end: the Riemannian geometry of $\text{SDiff}(M^3)$ beyond its conserved quantities. On the family of Question 7.2 it separates $s = 0$ from the rest. The exponential map is a nonlinear Fredholm map for positive integer orders [11] and is not Fredholm at $s = 0$ in three dimensions [8], and that failure is what makes conjugate points there a phenomenon local to single particle paths [13, 14]. This gives a geometric reason, alongside the scaling count, why nothing established along the dial transfers to Euler [18]. At $s = 0$ the local theory yields a dichotomy for blowup [15]. Suppose the vorticity integrates to infinity along a particle path. Then either the geodesic stops minimizing on successive intervals accumulating at the blowup time, so that the sectional curvature reaches $+\infty$, or the eigenvectors of the stretching matrix $D\eta^\top D\eta$ align in fixed directions, a regime compatible with negative curvature. At the fixed ring on the boundary of a cylinder, swirl odd in the axial coordinate rules out curve-shortening variations localized near the ring, and drives the pressure Laplacian and the sectional curvature to $-\infty$ in the time-integrated sense [16, Thm. 1.5]. That is the configuration in which finite-time blowup from smooth data has since been proved [7]. The Riemannian structure therefore classifies blowup without excluding it.

Read together with [20], which works without a boundary and gives up the exact metric, this locates what remains: the exact L^2 metric on a manifold without boundary. There the pressure Hessian plays the role of the curvature, through the Gauss–Codazzi formula for the sectional curvature [16, (24)], and it is produced by the periodic elliptic problem $\Delta p = -\text{tr}(\nabla u)^2$. We have no tool that reaches it. The conditions that a quantity reaching it must satisfy are collected in [22].

Theorem 6.3 assumes A is translation covariant; the general case is covered only by the blunter Theorem 6.2, as discussed in Remark 6.7. The Lie bracket is fixed throughout, so all statements are of the form “not Euler–Arnold for this bracket with any inertia operator”. The hypotheses of Theorems 3.8, 5.2 and 5.9 — reachability (Definition 3.7), transversality (Definition 5.8), and connectedness of the triangle and triad graphs — are finite conditions on a

symbol or on a truncation. We have not shown that they hold in general, but we have verified that they hold in every case used in this paper, and we record the outcome explicitly:

- *Reachability* holds in all 25 cases $N \in \{5, 7, 9, 11, 13\}$ against each of the five symbols of Remark 3.10; in each case the closure of Definition 3.7 exhausts Λ , starting from $\{(0, 1), (1, 1)\}$.
- The *triangle graph* is connected, with a single component, for every $N \in \{5, 7, 9, 11, 13\}$.
- The *triad graph* is connected, with a single component, for $\{|k| \leq K\} \subset \mathbb{Z}^3$ with $K = 2, 3, 4$ (respectively 32, 122 and 256 modes, and 68, 1092 and 5032 non-degenerate triads).
- *Transversality* holds for $K = 2, 3, 4$: no mode fails to lie in two triads with $\nu_1 \cdot \nu_2 \notin \{0, \pm 1\}$, out of 32, 122 and 256 modes respectively.

Consequently Theorems 3.8, 5.2 and 5.9 apply, as stated, to every case considered here. The script `verify_hypotheses.py` accompanying this paper rechecks all four families, reports each condition separately, and exits non-zero on any failure.

Nothing here bears on the regularity of the Navier–Stokes equation itself.

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The code underlying the computational statements is available from the author.

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